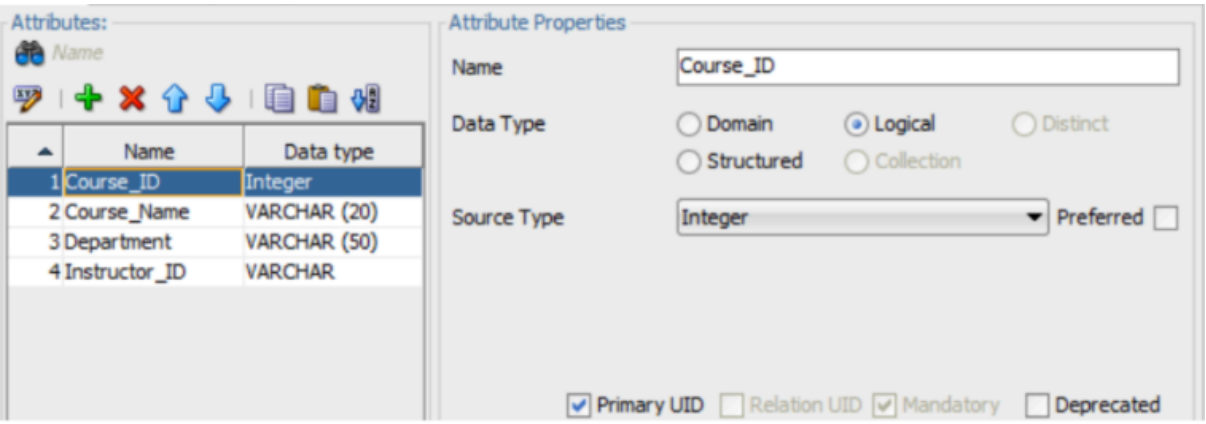
Data Base System

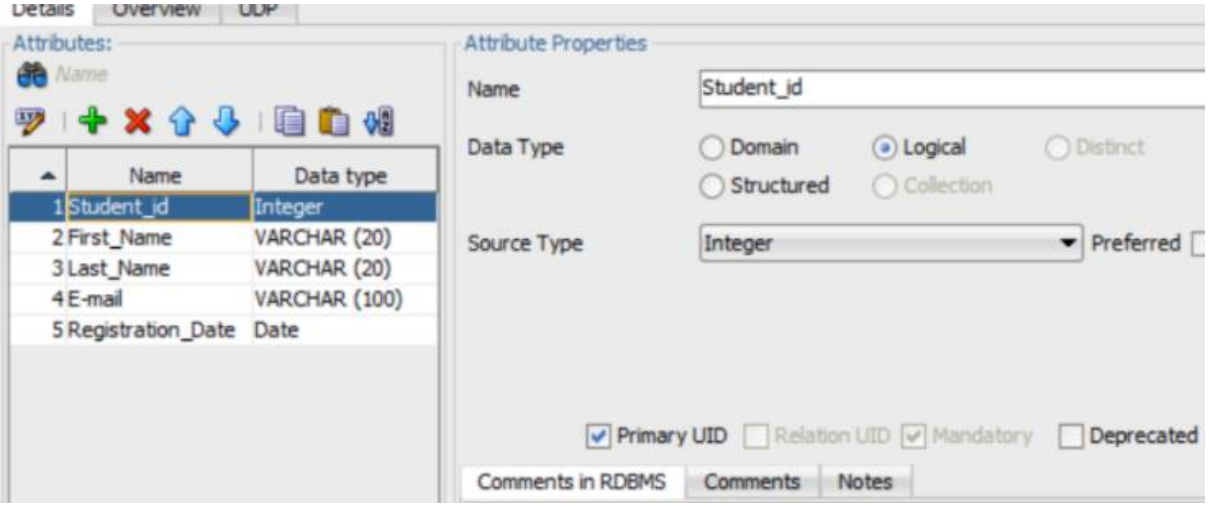
Muhammad Adil Saeed | 24k-0705 | Lab 9

In Lab Task

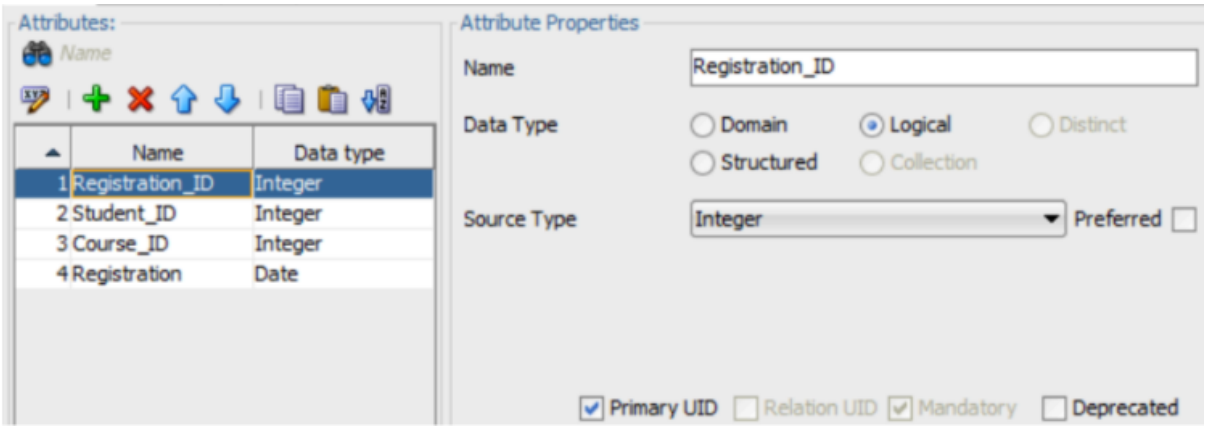
1.



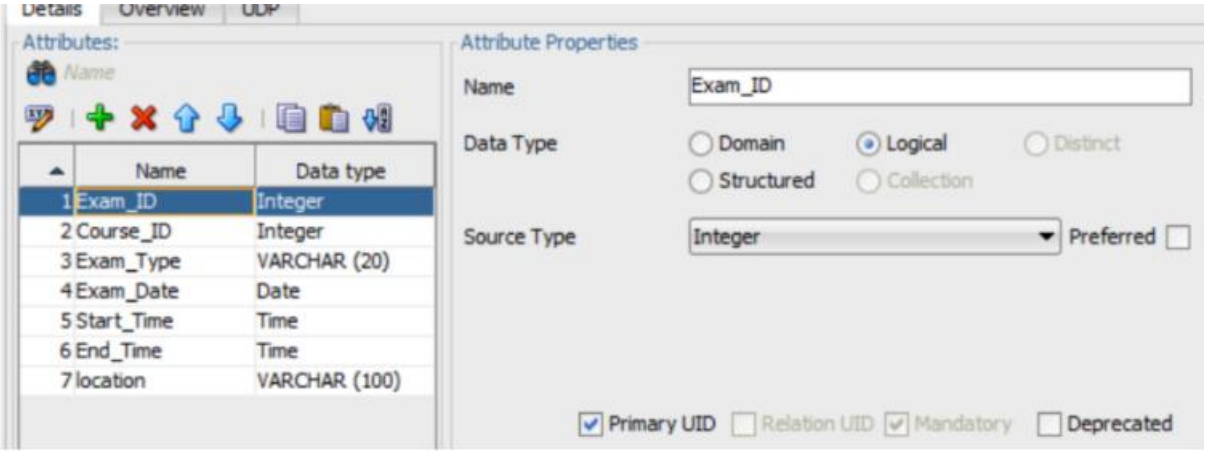
2.



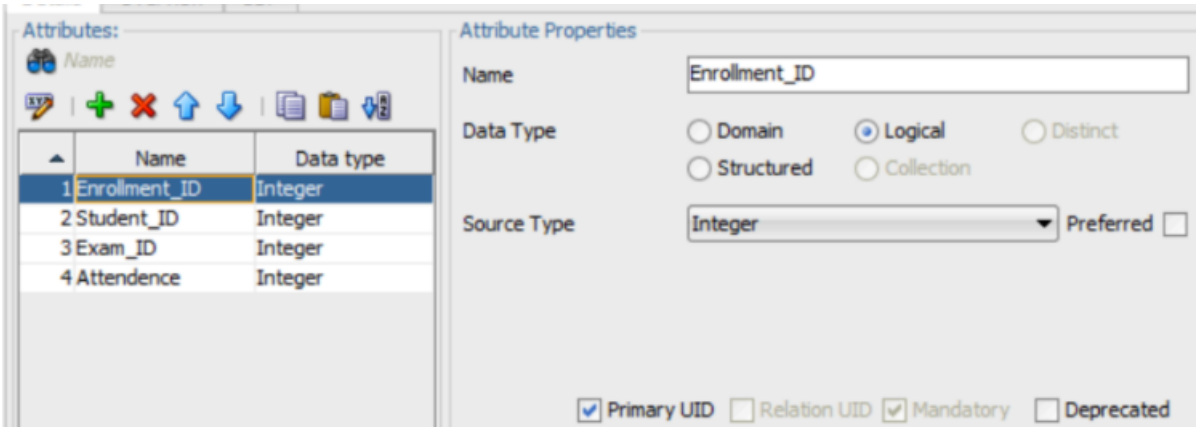
3.



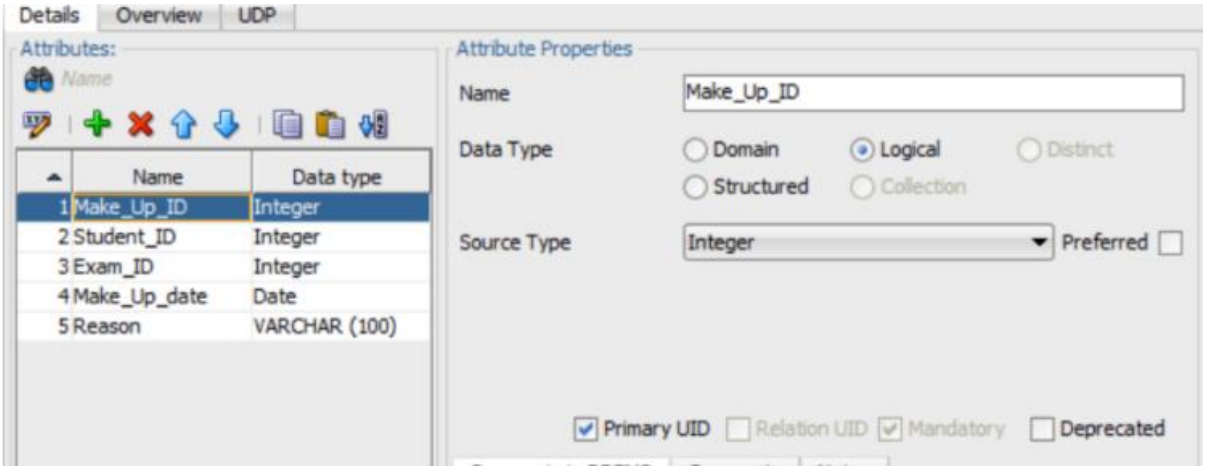
4.


Details
Overview
UDP
Attributes:
Name
Name
Data type
1 Exam_ID
Integer
2 Course_ID
Integer
3 Exam_Type
VARCHAR (20)
4 Exam_Date
Date
5 Start_Time
Time
6 End_Time
Time
7 location
VARCHAR (100)
Attribute Properties
Name
Exam_ID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

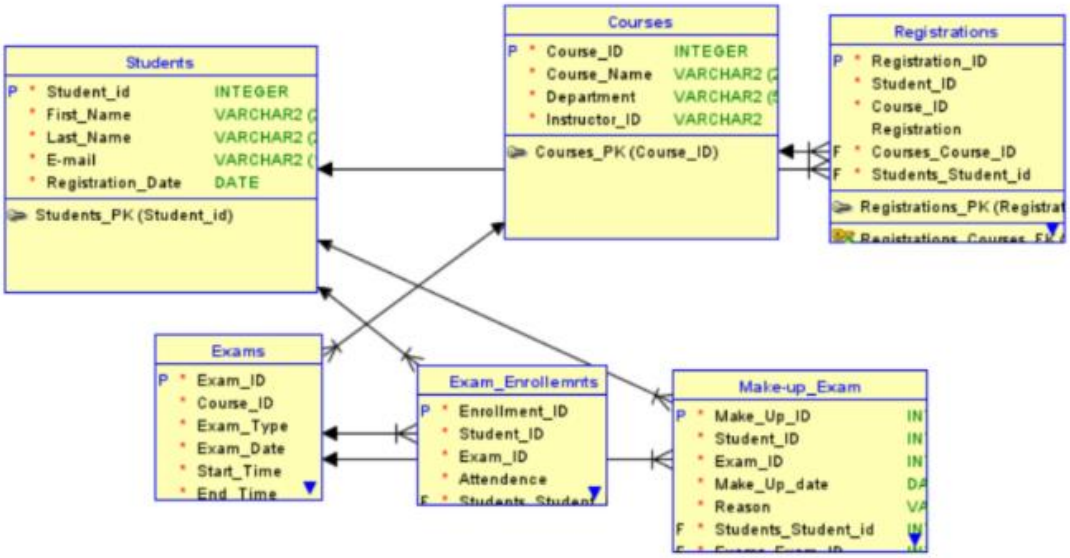
5.


Attributes:
Name
Name
Data type
1 Enrollment_ID
Integer
2 Student_ID
Integer
3 Exam_ID
Integer
4 Attendence
Integer
Attribute Properties
Name
Enrollment_ID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

6.


Details
Overview
UDP
Attributes:
Name
Name
Data type
1 Make_Up_ID
Integer
2 Student_ID
Integer
3 Exam_ID
Integer
4 Make_Up_date
Date
5 Reason
VARCHAR (100)
Attribute Properties
Name
Make_Up_ID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

7.



8.

```
-- Generated by Oracle SQL Developer Data Modeler 24.3.1.351.0831
--   at:        2026-04-04 20:16:13 PKT
--   site:      Oracle Database 11g
--   type:      Oracle Database 11g



-- predefined type, no DDL - MDSYS.SDO_GEOMETRY

-- predefined type, no DDL - XMLTYPE

CREATE TABLE Courses
    (
     Course_ID     INTEGER  NOT NULL ,
     Course_Name   VARCHAR2 (20)  NOT NULL ,
     Department    VARCHAR2 (50)  NOT NULL ,
     Instructor_ID VARCHAR2
--   ERROR: VARCHAR2 size not specified
                   NOT NULL
    )
;

ALTER TABLE Courses
    ADD CONSTRAINT Courses_PK PRIMARY KEY ( Course_ID ) ;
```

```
CREATE TABLE Exam_Enrollemnts
    (
     Enrollment_ID       INTEGER  NOT NULL ,
     Student_ID          INTEGER  NOT NULL ,
     Exam_ID             INTEGER  NOT NULL ,
     Attendence          INTEGER  NOT NULL ,
     Students_Student_id INTEGER  NOT NULL ,
     Exams_Exam_ID       INTEGER  NOT NULL
    )
;

ALTER TABLE Exam_Enrollemnts
    ADD CONSTRAINT Exam_Enrollemnts_PK PRIMARY KEY ( Enrollment_ID ) ;

CREATE TABLE Exams
    (
     Exam_ID           INTEGER  NOT NULL ,
     Course_ID         INTEGER  NOT NULL ,
```

```
CREATE TABLE Exams
    (
     Exam_ID           INTEGER  NOT NULL ,
     Course_ID         INTEGER  NOT NULL ,
     Exam_Type         VARCHAR2 (20)  NOT NULL ,
     Exam_Date         DATE  NOT NULL ,
     Start_Time        DATE  NOT NULL ,
     End_Time          DATE  NOT NULL ,
     location          VARCHAR2 (100)  NOT NULL ,
     Courses_Course_ID INTEGER  NOT NULL
    )
;

ALTER TABLE Exams
    ADD CONSTRAINT Exams_PK PRIMARY KEY ( Exam_ID ) ;

CREATE TABLE "Make-up_Exam"
    (
     Make_Up_ID          INTEGER  NOT NULL ,
     Student_ID          INTEGER  NOT NULL ,
     Exam_ID             INTEGER  NOT NULL ,
     Make_Up_date        DATE  NOT NULL ,
     Reason              VARCHAR2 (100)  NOT NULL ,
     Students_Student_id INTEGER  NOT NULL ,
     Exams_Exam_ID       INTEGER  NOT NULL
    )
;
```

```
CREATE TABLE "Make-up_Exam"
    (
     Make_Up_ID          INTEGER  NOT NULL ,
     Student_ID          INTEGER  NOT NULL ,
     Exam_ID             INTEGER  NOT NULL ,
     Make_Up_date        DATE  NOT NULL ,
     Reason              VARCHAR2 (100)  NOT NULL ,
     Students_Student_id INTEGER  NOT NULL ,
     Exams_Exam_ID       INTEGER  NOT NULL
    )
;

ALTER TABLE "Make-up_Exam"
    ADD CONSTRAINT "Make-up_Exam_PK" PRIMARY KEY ( Make_Up_ID ) ;

CREATE TABLE Registrations
    (
     Registration_ID     INTEGER  NOT NULL ,
     Student_ID          INTEGER  NOT NULL ,
     Course_ID           INTEGER  NOT NULL ,
     Registration        DATE ,
     Courses_Course_ID   INTEGER  NOT NULL ,
     Students_Student_id INTEGER  NOT NULL
    )
;
```

```sql
ALTER TABLE Registrations
    ADD CONSTRAINT Registrations_PK PRIMARY KEY ( Registration_ID ) ;

CREATE TABLE Students
    (
     Student_id        INTEGER  NOT NULL ,
     First_Name        VARCHAR2 (20)  NOT NULL ,
     Last_Name         VARCHAR2 (20)  NOT NULL ,
     "E-mail"          VARCHAR2 (100)  NOT NULL ,
     Registration_Date DATE  NOT NULL
    )
;

ALTER TABLE Students
    ADD CONSTRAINT Students_PK PRIMARY KEY ( Student_id ) ;

ALTER TABLE Exam_Enrollemnts
    ADD CONSTRAINT Exam_Enrollemnts_Exams_FK FOREIGN KEY
    (
     Exams_Exam_ID
    )
    REFERENCES Exams
    (
     Exam_ID
    )
;

ALTER TABLE Exam_Enrollemnts
    ADD CONSTRAINT Exam_Enrollemnts_Students_FK FOREIGN KEY
    (
     Students_Student_id
    )
    REFERENCES Students
    (
     Student_id
    )
;
```

```sql
ALTER TABLE Exams
    ADD CONSTRAINT Exams_Courses_FK FOREIGN KEY
    (
     Courses_Course_ID
    )
    REFERENCES Courses
    (
     Course_ID
    )
;

ALTER TABLE "Make-up_Exam"
    ADD CONSTRAINT "Make-up_Exam_Exams_FK" FOREIGN KEY
    (
     Exams_Exam_ID
    )
    REFERENCES Exams
    (
     Exam_ID
    )
;
```

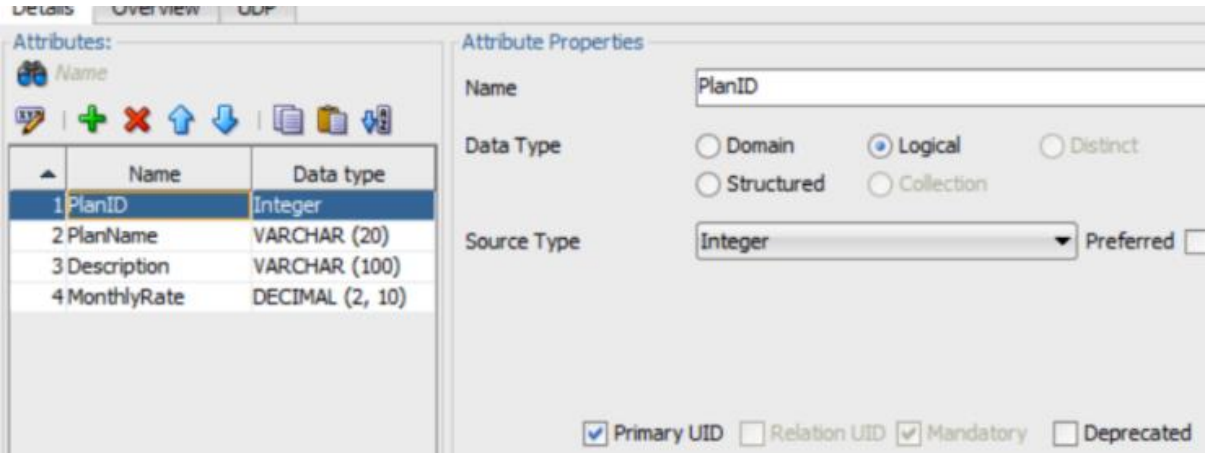
```sql
ALTER TABLE "Make-up_Exam"
    ADD CONSTRAINT "Make-up_Exam_Students_FK" FOREIGN KEY
    (
     Students_Student_id
    )
    REFERENCES Students
    (
     Student_id
    )
;

ALTER TABLE Registrations
    ADD CONSTRAINT Registrations_Courses_FK FOREIGN KEY
    (
     Courses_Course_ID
    )
    REFERENCES Courses
    (
     Course_ID
    )
;
```

```sql
ALTER TABLE Registrations
    ADD CONSTRAINT Registrations_Students_FK FOREIGN KEY
    (
     Students_Student_id
    )
    REFERENCES Students
    (
     Student_id
    )
;
```

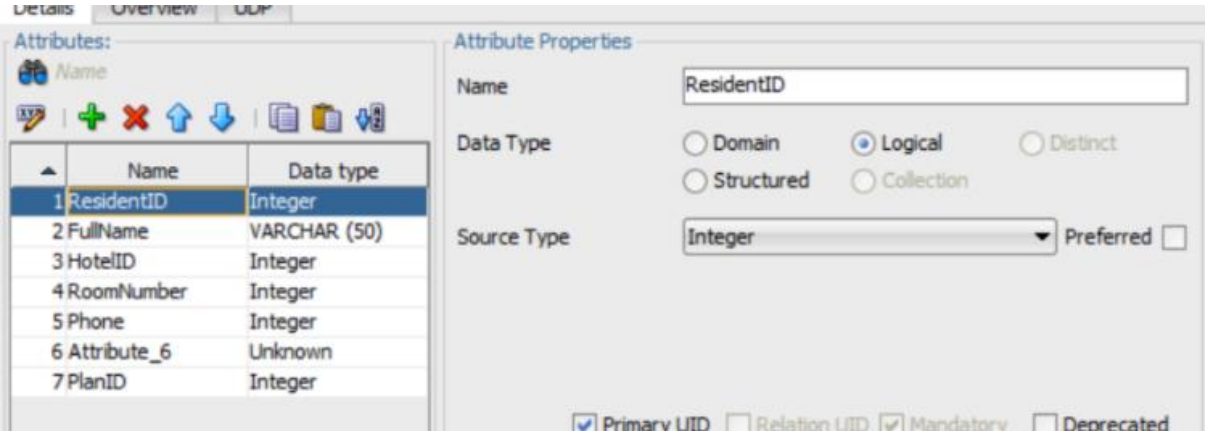
```
-- Oracle SQL Developer Data Modeler Summary Report:
--
-- CREATE TABLE                             6
-- CREATE INDEX                             0
-- ALTER TABLE                             13
-- CREATE VIEW                              0
-- ALTER VIEW                               0
-- CREATE PACKAGE                           0
-- CREATE PACKAGE BODY                      0
-- CREATE PROCEDURE                         0
-- CREATE FUNCTION                          0
-- CREATE TRIGGER                           0
-- ALTER TRIGGER                            0
-- CREATE COLLECTION TYPE                   0
-- CREATE STRUCTURED TYPE                   0
-- CREATE STRUCTURED TYPE BODY              0
-- CREATE CLUSTER                           0
-- CREATE CONTEXT                           0
-- CREATE DATABASE                          0
-- CREATE DIMENSION                         0
-- CREATE DIRECTORY                         0
-- CREATE DISK GROUP                        0
-- CREATE ROLE                              0
-- CREATE ROLLBACK SEGMENT                  0
-- CREATE SEQUENCE                          0
-- CREATE MATERIALIZED VIEW                 0
-- CREATE MATERIALIZED VIEW LOG             0
-- CREATE SYNONYM                           0
-- CREATE TABLESPACE                        0
```

Post Lab Task

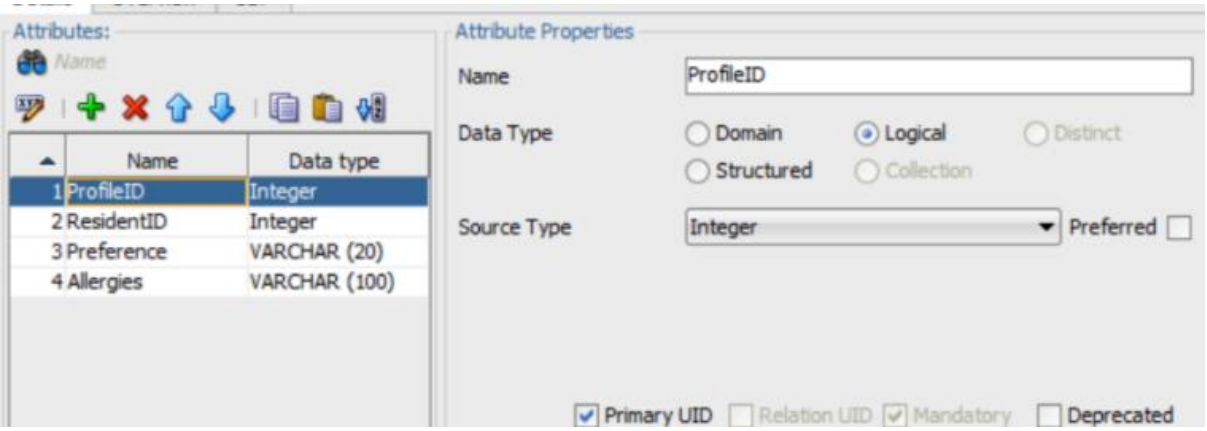
1.



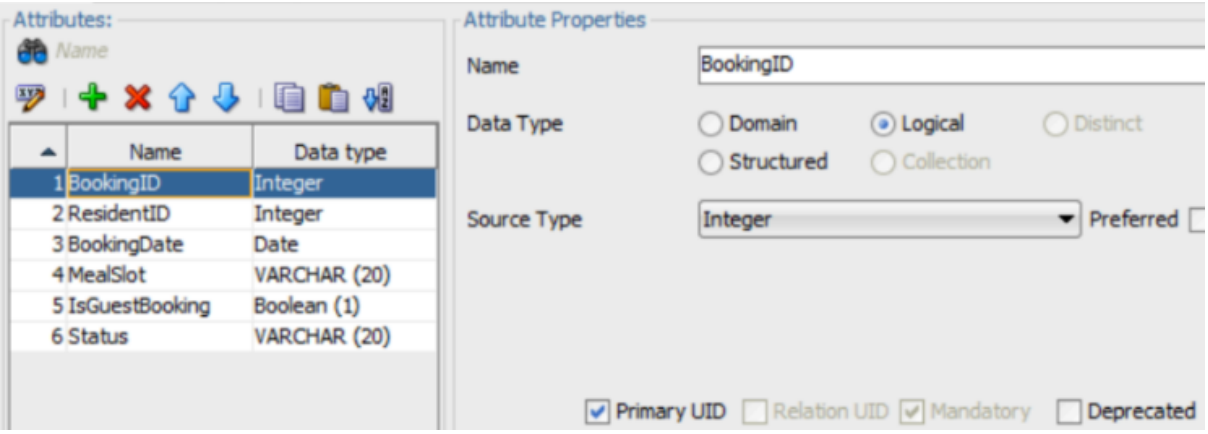
2.



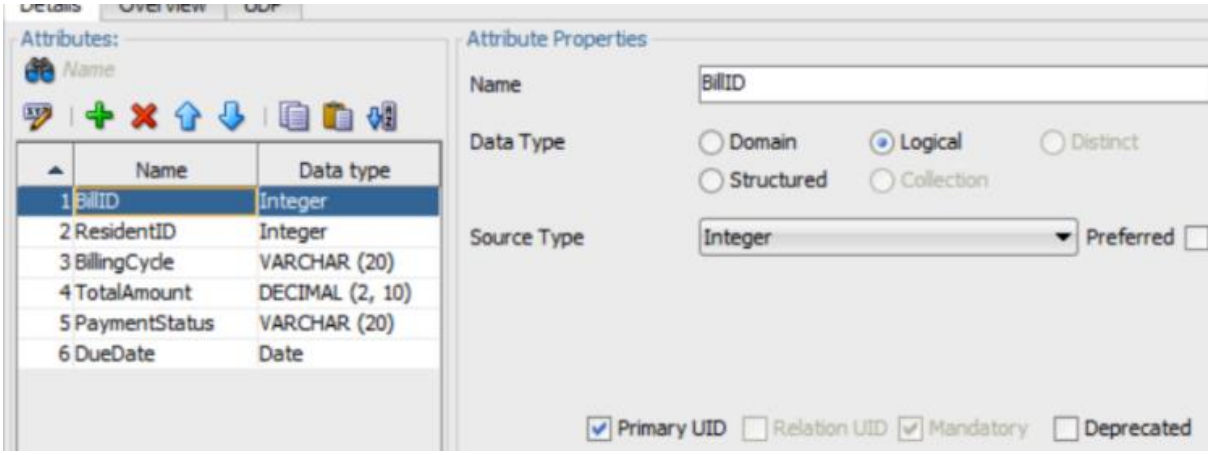
3.



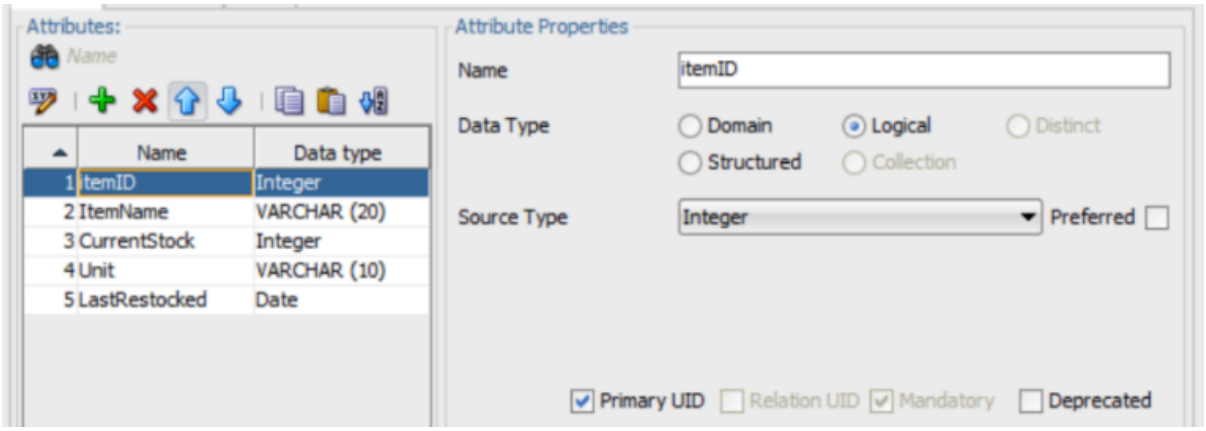
4.


Attributes:
Name
Name
Data type
1 BookingID
Integer
2 ResidentID
Integer
3 BookingDate
Date
4 MealSlot
VARCHAR (20)
5 IsGuestBooking
Boolean (1)
6 Status
VARCHAR (20)
Attribute Properties
Name
BookingID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

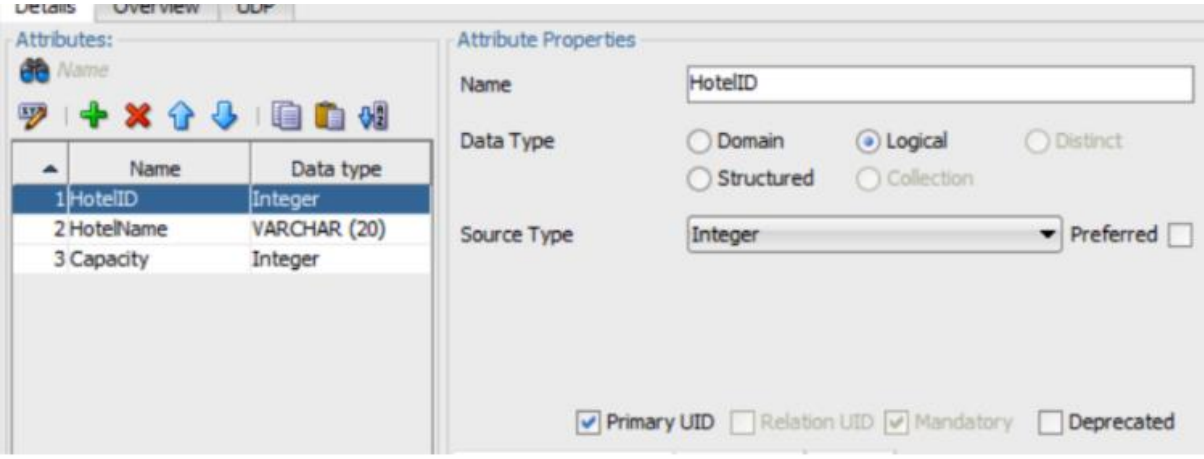
5.


Attributes:
Name
Name
Data type
1 BillID
Integer
2 ResidentID
Integer
3 BillingCycle
VARCHAR (20)
4 TotalAmount
DECIMAL (2, 10)
5 PaymentStatus
VARCHAR (20)
6 DueDate
Date
Attribute Properties
Name
BillID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

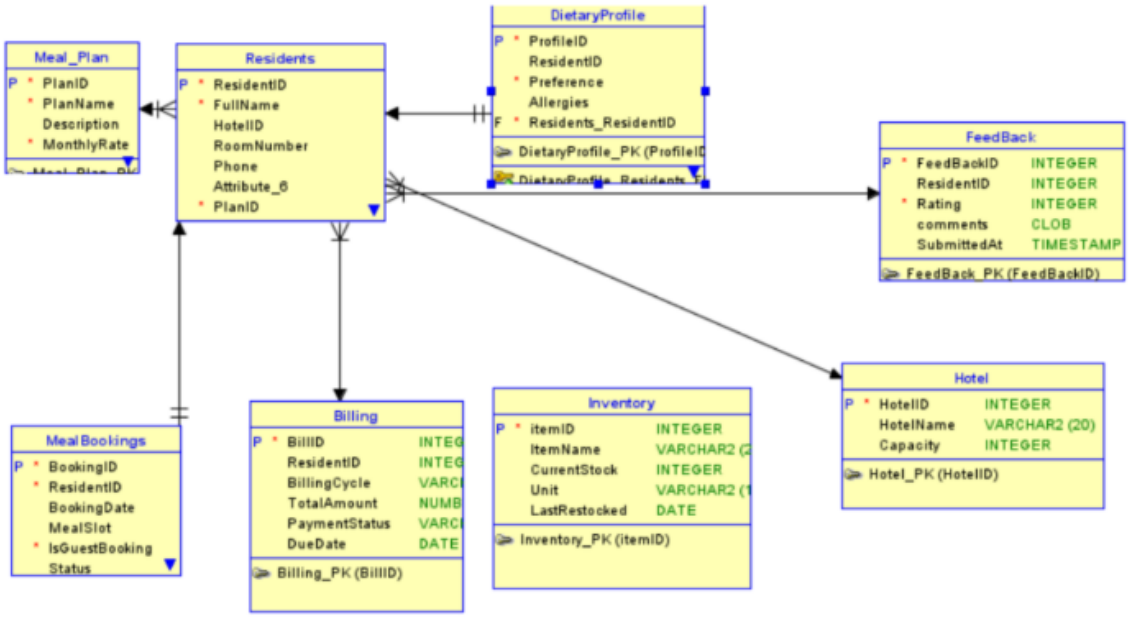
6.


Attributes:
Name
Name
Data type
1 itemID
Integer
2 ItemName
VARCHAR (20)
3 CurrentStock
Integer
4 Unit
VARCHAR (10)
5 LastRestocked
Date
Attribute Properties
Name
itemID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

7.


Details
Overview
UDP
Attributes:
Name
Name
Data type
1 HotelID
Integer
2 HotelName
VARCHAR (20)
3 Capacity
Integer
Attribute Properties
Name
HotelID
Data Type
Domain
Logical
Distinct
Structured
Collection
Source Type
Integer
Preferred
Primary UID
Relation UID
Mandatory
Deprecated

8.


Meal_Plan
PlanID
PlanName
Description
MonthlyRate
Residents
ResidentID
FullName
HotelID
RoomNumber
Phone
Attribute_6
PlanID
DietaryProfile
ProfileID
ResidentID
Preference
Allergies
Residents_ResidentID
DietaryProfile_PK (ProfileID
FeedBack
FeedBackID INTEGER
ResidentID INTEGER
Rating INTEGER
comments CLOB
SubmittedAt TIMESTAMP
FeedBack_PK (FeedBackID)
Hotel
HotelID INTEGER
HotelName VARCHAR2 (20)
Capacity INTEGER
Hotel_PK (HotelID)
Inventory
itemID INTEGER
ItemName VARCHAR2
CurrentStock INTEGER
Unit VARCHAR2
LastRestocked DATE
Inventory_PK (itemID)
Billing
BillID
ResidentID
BillingCycle
TotalAmount
PaymentStatus
DueDate DATE
Billing_PK (BillID)
MealBookings
BookingID
ResidentID
BookingDate
MealSlot
IsGuestBooking
Status

9.

```sql
-- Generated by Oracle SQL Developer Data Modeler 24.3.1.351.0831
--   at:        2026-04-04 22:09:17 PKT
--   site:      Oracle Database 11g
--   type:      Oracle Database 11g



-- predefined type, no DDL - MDSYS.SDO_GEOMETRY

-- predefined type, no DDL - XMLTYPE

CREATE TABLE Billing
    (
     BillID        INTEGER  NOT NULL ,
     ResidentID    INTEGER ,
     BillingCycle  VARCHAR2 (20) ,
     TotalAmount   NUMBER (2,10) ,
     PaymentStatus VARCHAR2 (20) ,
     DueDate       DATE
    )
;

ALTER TABLE Billing
    ADD CONSTRAINT Billing_PK PRIMARY KEY ( BillID ) ;

CREATE TABLE DietaryProfile
    (
     ProfileID            INTEGER  NOT NULL ,
     ResidentID           INTEGER ,
     Preference           VARCHAR2 (20)  NOT NULL ,
     Allergies            VARCHAR2 (100) ,
     Residents_ResidentID INTEGER  NOT NULL
    )
;
```

```
CREATE TABLE DietaryProfile
    (
     ProfileID            INTEGER  NOT NULL ,
     ResidentID           INTEGER ,
     Preference           VARCHAR2 (20)  NOT NULL ,
     Allergies            VARCHAR2 (100) ,
     Residents_ResidentID INTEGER  NOT NULL
    )
;
CREATE UNIQUE INDEX DietaryProfile__IDX ON DietaryProfile
    (
     Residents_ResidentID ASC
    )
;

ALTER TABLE DietaryProfile
    ADD CONSTRAINT DietaryProfile_PK PRIMARY KEY ( ProfileID ) ;

CREATE TABLE FeedBack
    (
     FeedBackID  INTEGER  NOT NULL ,
     ResidentID  INTEGER ,
     Rating      INTEGER  NOT NULL ,
     comments    CLOB ,
     SubmittedAt TIMESTAMP
    )
;

ALTER TABLE FeedBack
    ADD CONSTRAINT FeedBack_PK PRIMARY KEY ( FeedBackID ) ;

CREATE TABLE Hotel
    (
     HotelID   INTEGER  NOT NULL ,
     HotelName VARCHAR2 (20) ,
     Capacity  INTEGER
    )
;

ALTER TABLE Hotel
    ADD CONSTRAINT Hotel_PK PRIMARY KEY ( HotelID ) ;
```

```sql
CREATE TABLE Inventory
    (
     itemID        INTEGER  NOT NULL ,
     ItemName      VARCHAR2 (20) ,
     CurrentStock  INTEGER ,
     Unit          VARCHAR2 (10) ,
     LastRestocked DATE
    )
;

ALTER TABLE Inventory
    ADD CONSTRAINT Inventory_PK PRIMARY KEY ( itemID ) ;

CREATE TABLE Meal_Plan
    (
     PlanID      INTEGER  NOT NULL ,
     PlanName    VARCHAR2 (20)  NOT NULL ,
     Description VARCHAR2 (100) ,
     MonthlyRate NUMBER (2,10)  NOT NULL
    )
;

ALTER TABLE Meal_Plan
    ADD CONSTRAINT Meal_Plan_PK PRIMARY KEY ( PlanID ) ;

CREATE TABLE MealBookings
    (
     BookingID            INTEGER  NOT NULL ,
     ResidentID           INTEGER  NOT NULL ,
     BookingDate          DATE ,
     MealSlot             VARCHAR2 (20) ,
     IsGuestBooking       CHAR (1)  NOT NULL ,
     Status               VARCHAR2 (20) ,
     Residents_ResidentID INTEGER  NOT NULL
    )
;
```

```
CREATE UNIQUE INDEX MealBookings__IDX ON MealBookings
    (
     Residents_ResidentID ASC
    )
;

ALTER TABLE MealBookings
    ADD CONSTRAINT MealBookings_PK PRIMARY KEY ( BookingID ) ;

CREATE TABLE Residents
    (
     ResidentID          INTEGER  NOT NULL ,
     FullName            VARCHAR2 (50)  NOT NULL ,
     HotelID             INTEGER ,
     RoomNumber          INTEGER ,
     Phone               INTEGER ,
     PlanID              INTEGER  NOT NULL ,
     Hotel_HotelID       INTEGER  NOT NULL ,
     Meal_Plan_PlanID    INTEGER  NOT NULL ,
     Billing_BillID      INTEGER  NOT NULL ,
     FeedBack_FeedBackID INTEGER  NOT NULL
    )
;

ALTER TABLE Residents
    ADD CONSTRAINT Residents_PK PRIMARY KEY ( ResidentID ) ;
```

```sql
ALTER TABLE DietaryProfile
    ADD CONSTRAINT DietaryProfile_Residents_FK FOREIGN KEY
    (
     Residents_ResidentID
    )
    REFERENCES Residents
    (
     ResidentID
    )
;

ALTER TABLE MealBookings
    ADD CONSTRAINT MealBookings_Residents_FK FOREIGN KEY
    (
     Residents_ResidentID
    )
    REFERENCES Residents
    (
     ResidentID
    )
;
```

```sql
ALTER TABLE Residents
    ADD CONSTRAINT Residents_Billing_FK FOREIGN KEY
    (
     Billing_BillID
    )
    REFERENCES Billing
    (
     BillID
    )
;

ALTER TABLE Residents
    ADD CONSTRAINT Residents_FeedBack_FK FOREIGN KEY
    (
     FeedBack_FeedBackID
    )
    REFERENCES FeedBack
    (
     FeedBackID
    )
;
```

```sql
ALTER TABLE Residents
    ADD CONSTRAINT Residents_Hotel_FK FOREIGN KEY
    (
     Hotel_HotelID
    )
    REFERENCES Hotel
    (
     HotelID
    )
;

ALTER TABLE Residents
    ADD CONSTRAINT Residents_Meal_Plan_FK FOREIGN KEY
    (
     Meal_Plan_PlanID
    )
    REFERENCES Meal_Plan
    (
     PlanID
    )
```